

Supplemental Material for:
Implementation of a minimal recurrent spiking neural network in a solid-state device

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1. Components

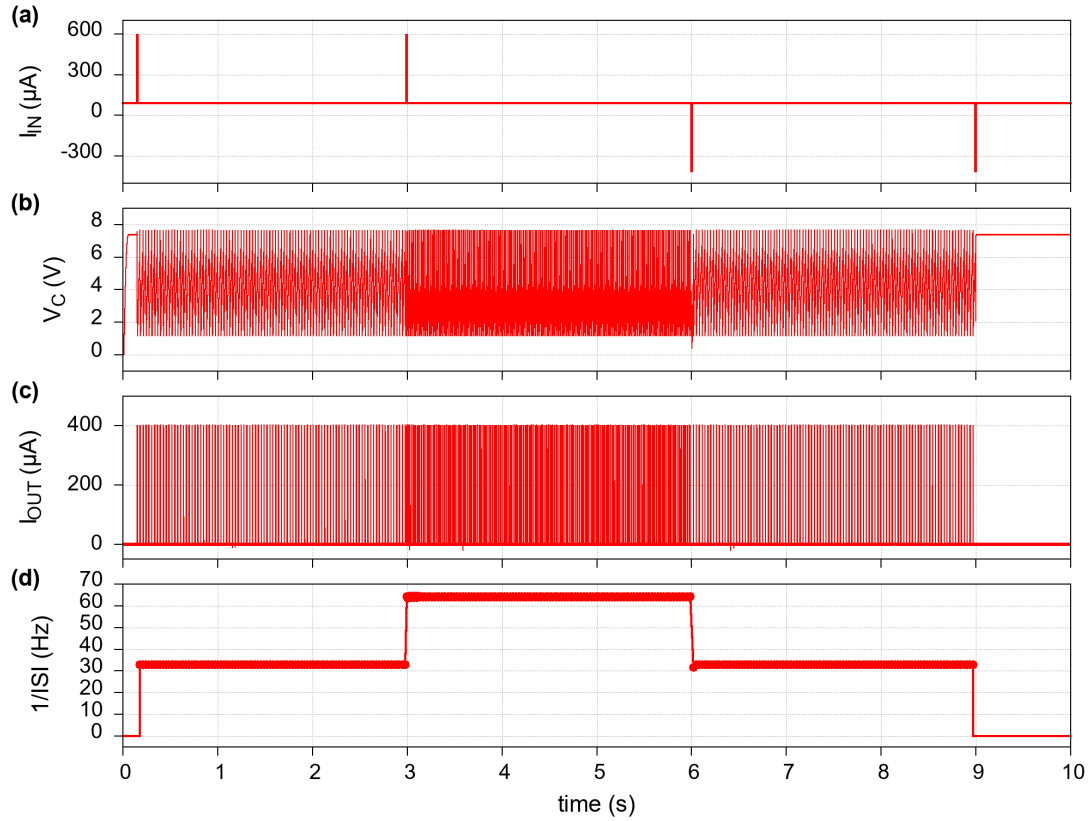
The table below provides the values of the components used in the circuit presented in Fig. 1 (Main Text)

Component	Value
C1	300nF
R1	270k Ω
R2	270k Ω
R3	2k Ω
R4	100k
R5	6.8k Ω
SCR	CR02A8
Q1	IRLML6246TR
Q2	ZVP3310A
Q3	ZVP3310A

Supplementary Table 1. List of components for the circuit in Fig.1 of the Main Text.

2. Autapse working memory

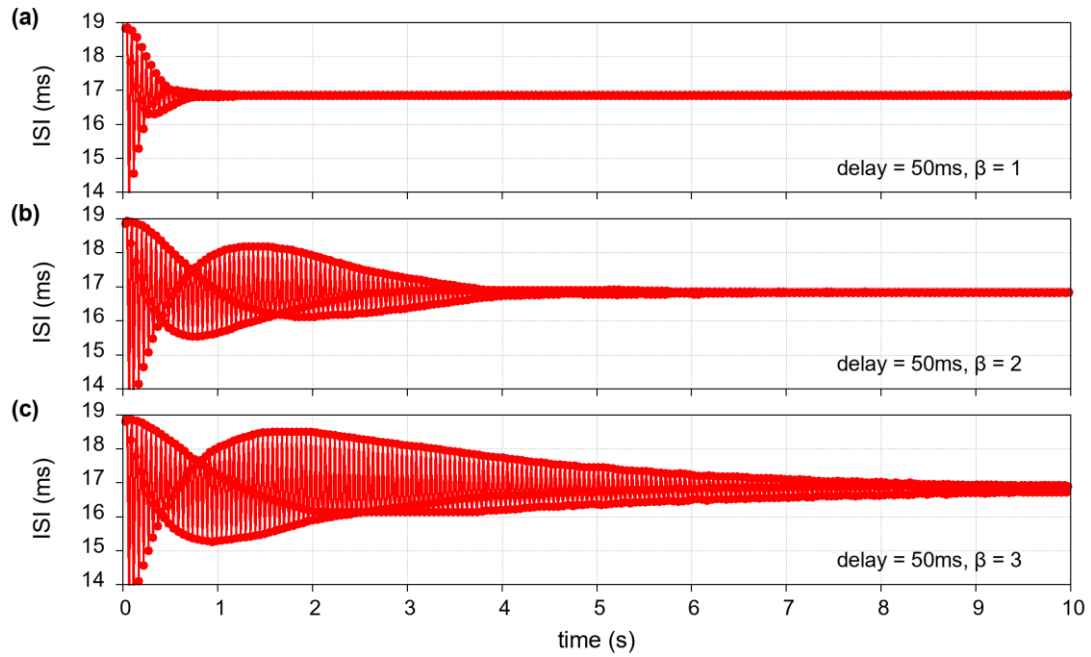
The figure below shows the behavior of the autapse working memory, that is the graded persistent activity states, during a long-time interval. It is equivalent to Fig. 5 of the main text, except that the individual spikes cannot be resolved due to the long duration of the interval. Notice that the spiking rates show no degradation.



Supplementary Fig. 1. Operation of the Autapse Memory during a long-time interval. (a) Input current to the UCNA. (b) Membrane voltage $V_C(t)$. (c) Output current. (d) Instantaneous spiking frequency.

3. Convergence Time

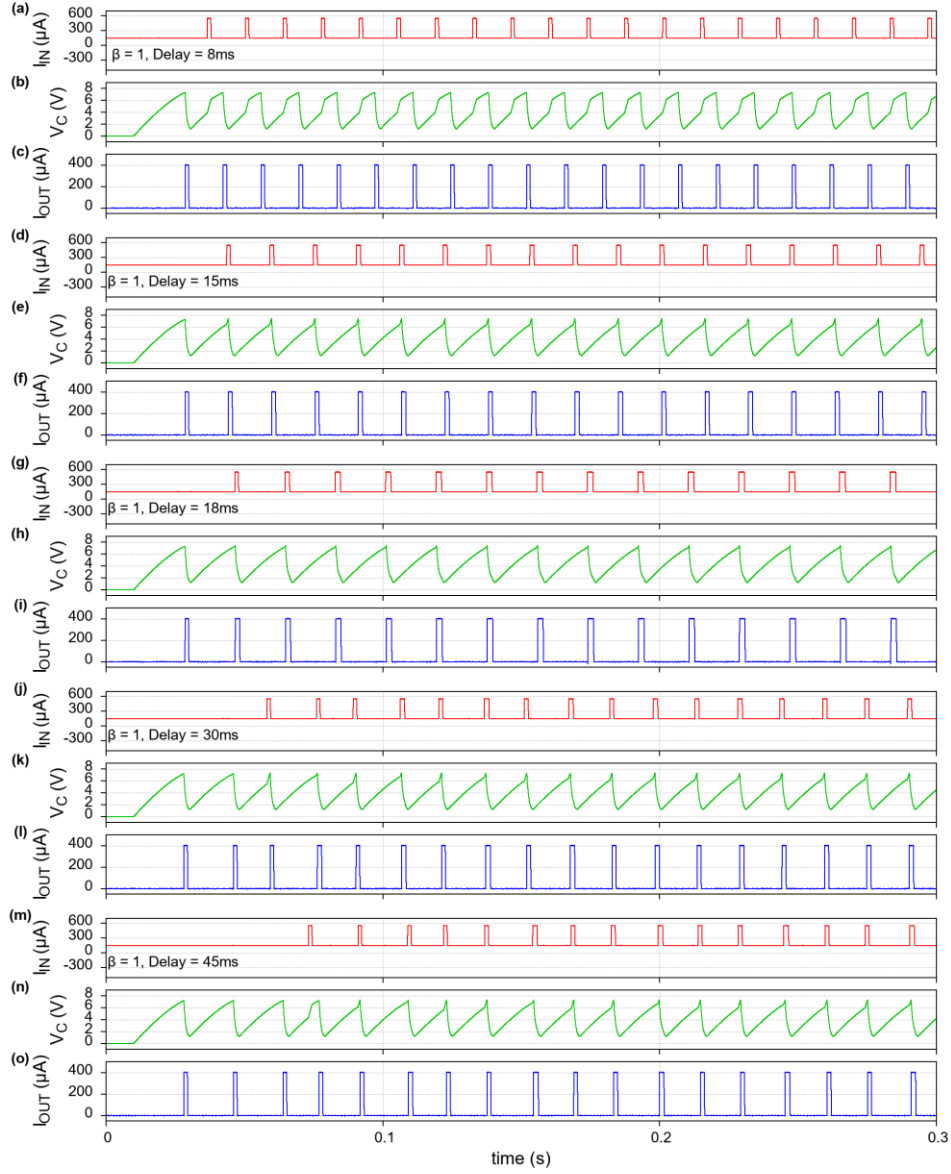
The Figure below show example of the time to reach to a self-consistent ISI for the UCNA.



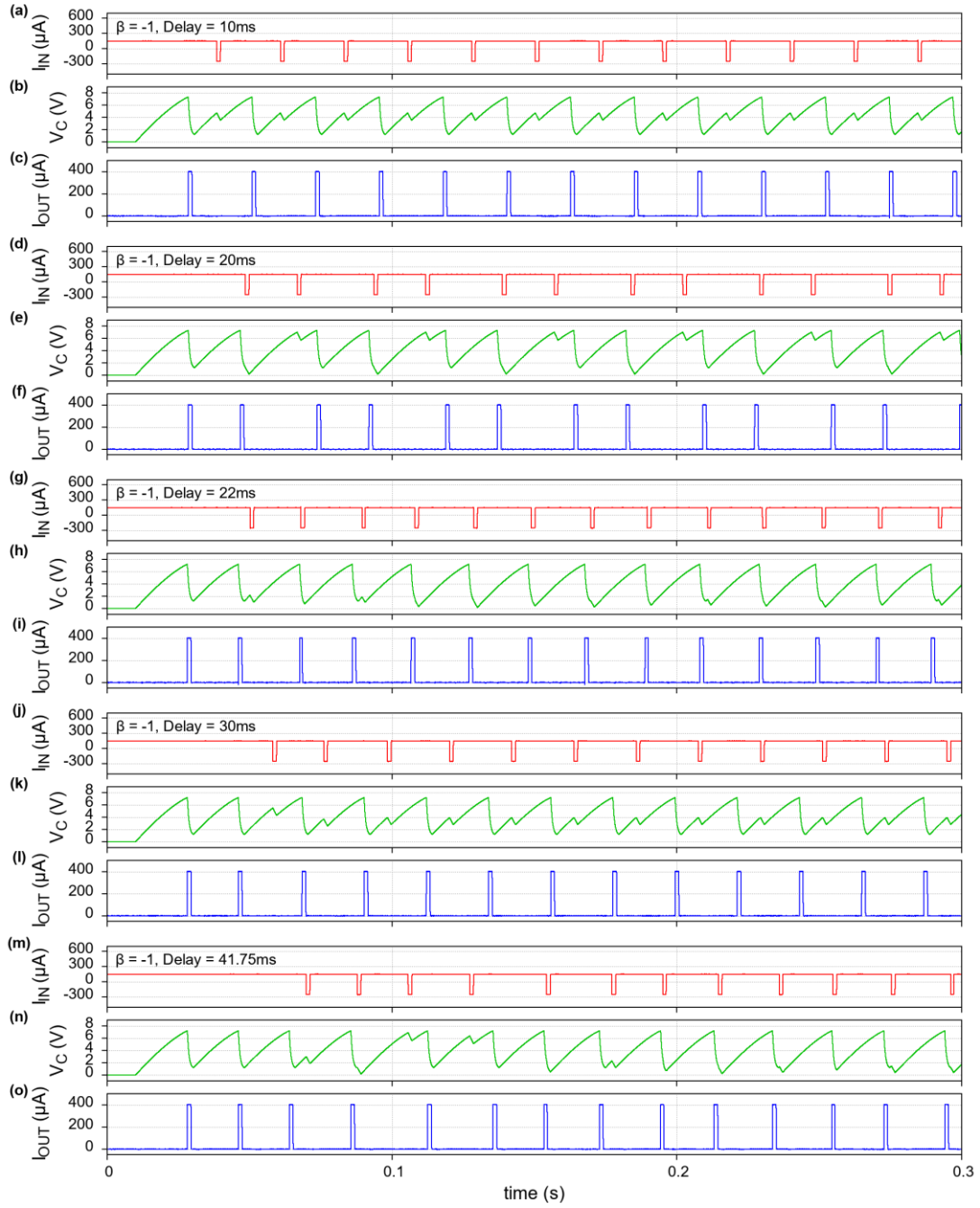
Supplementary Fig. 2. Convergence time of the UCNA. Panel (a) corresponds to the datapoint for delay=50ms of the Fig. 2(a) in the Main Text. Panels (b) and (c) are for increasingly bigger β values.

4. Additional waveforms

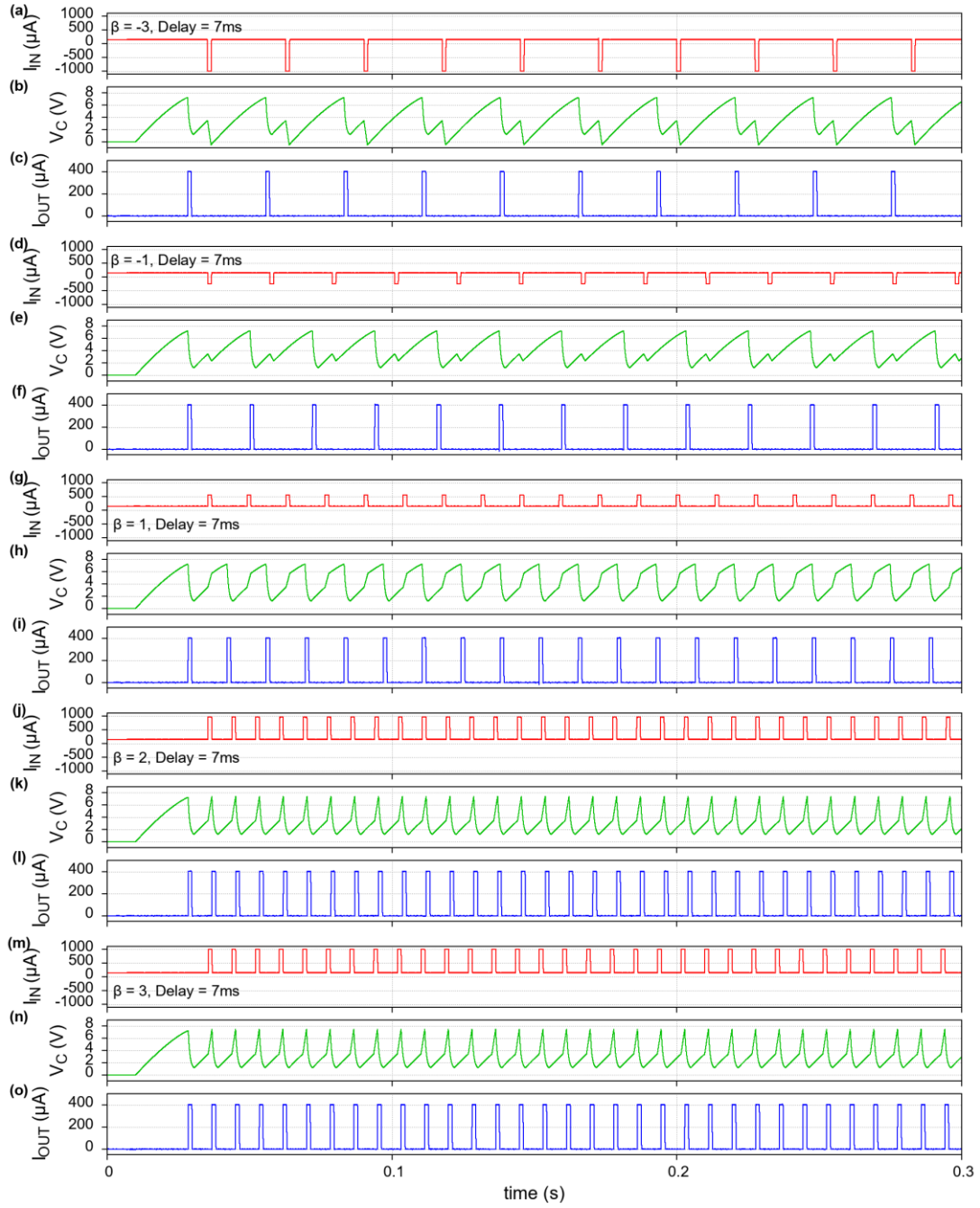
Below, we show measured waveforms, $I_{IN} = I_{IN0} + I_A$, V_C and I_{OUT} , for selected cases in Figs. 2 and Fig. 4 of the Main Text. Recall the autapse current $I_A(t) = \beta \cdot I_{OUT}(t - \Delta t)$.



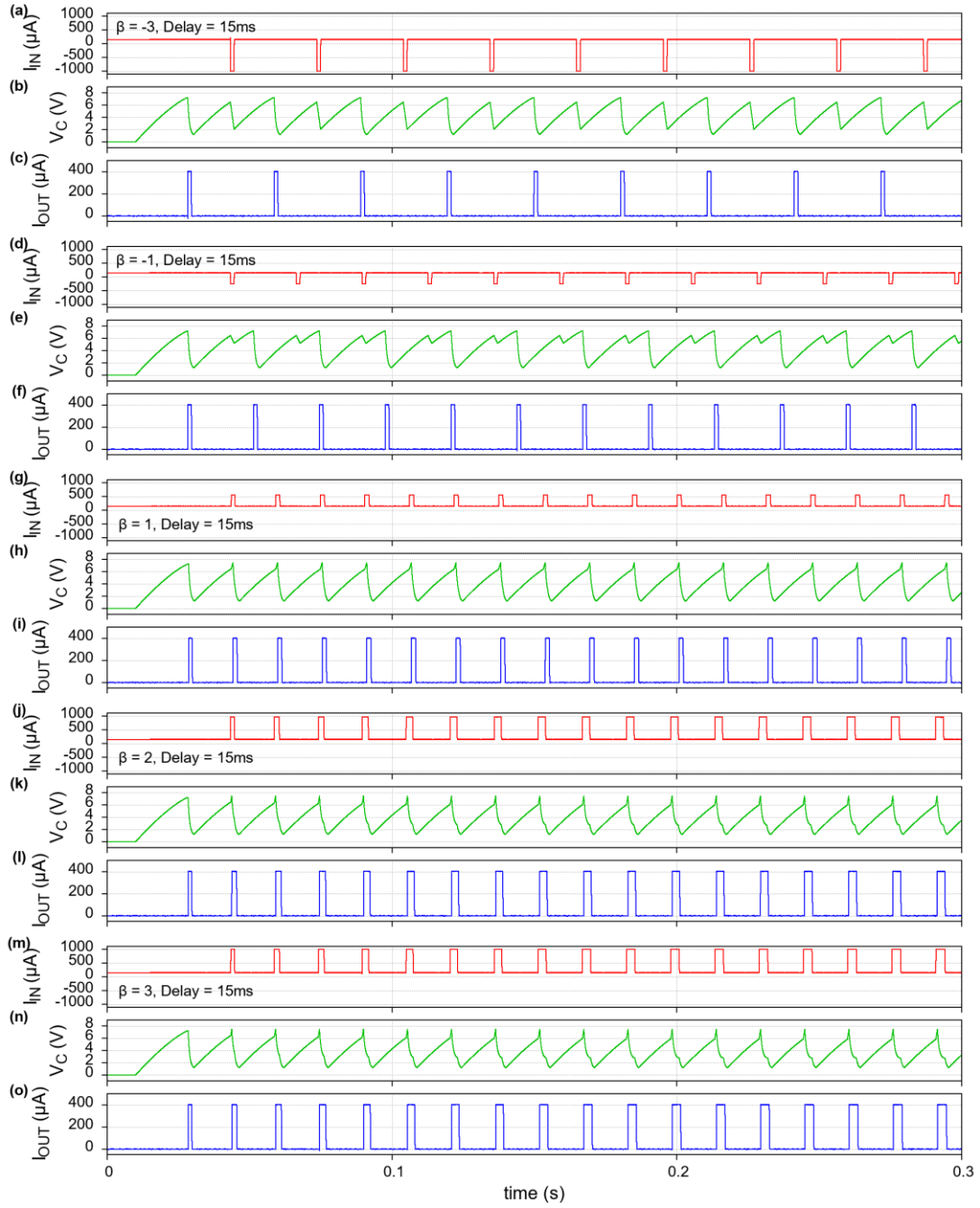
Supplementary Fig. 3. Detail of waveforms for selected excitatory cases in Fig. 2a in the Main Text. For the case $\beta = 1$, $\Delta t = 8\text{ms}$ we can observe how the spike contribution in I_{IN} suddenly brings the V_C from the linear charging regime to the non-linear one, resulting in the arch-like behavior in Fig. 2 of the Main Text. For the cases $\beta = 1$, $\Delta t = 15\text{ms}$ and $\beta = 1$, $\Delta t = 30\text{ms}$ we can clearly observe how the spike contribution in I_{IN} suddenly brings the V_C to the threshold for the fire discharge, reducing the ISI in the linear regime discussed in the Main Text. In contrast, in the case $\beta = 1$, $\Delta t = 18\text{ms}$ we can see how the spike contribution of I_{IN} is "wasted" as it arrives when the capacitor is discharging (i.e. during the firing), resulting in an increase of the ISI with respect to the ISI_0 .



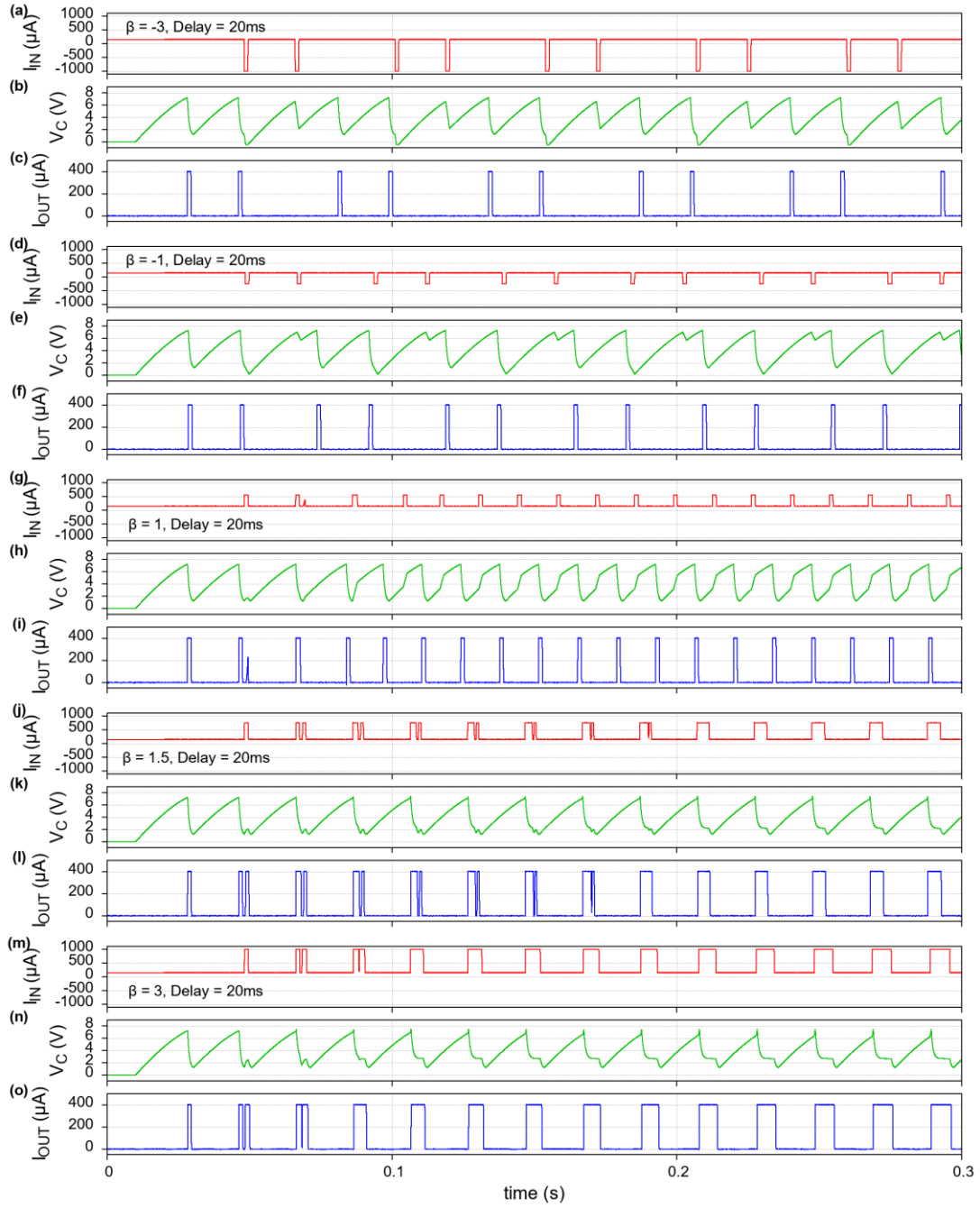
Supplementary Fig. 4. Detail of waveforms for selected inhibitory cases in Fig. 2b in the Main Text. Notice the persistent irregularities for $\Delta t = 20\text{ms}$, 22ms , 41.75ms , which correspond to open circles in the Fig. 2b.



Supplementary Fig. 5. Detail of waveforms for selected cases with $\Delta t = 7\text{ms}$ in Fig. 4a in the Main Text.



Supplementary Fig. 6. Detail of waveforms for selected cases with $\Delta t = 15\text{ms}$ in Fig. 4a in the Main Text.



Supplementary Fig. 7. Detail of waveforms for selected cases with $\Delta t = 20\text{ms}$ in Fig. 4b in the Main Text. Notice the persistent period-2 for the inhibitory cases at $\beta = -3$ and -1 .

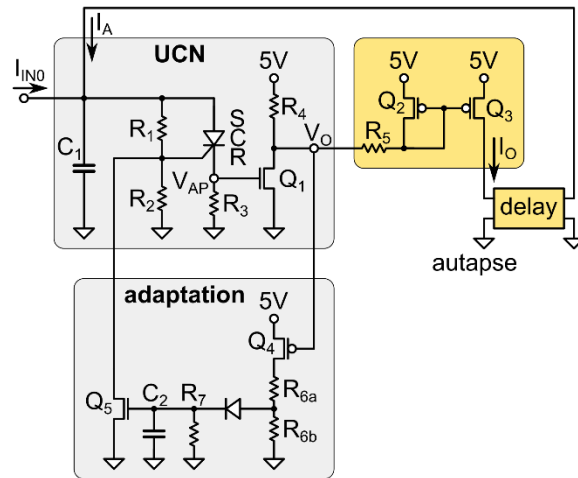
5. Autapse behavior in different neuron types

An interesting question is what is the effect of the recursion for spiking neuron types other than the leaky-integrate-and-fire (LIF) type considered in the main text. The existence of biological autapse forming in neurons with different spiking neuron patterns is not clear at this time. It is clearly a challenging experimental task to identify the "intrinsic" (ie with a switched off autapse feedback) neuron spiking type of biological neurons. Nevertheless, this question is also interesting from a neuromorphic engineering perspective as recursion may dramatically modify the spiking pattern and bring new insights in the effects of recursion for a spiking system, and perhaps even allow for novel functionalities.

This question is vast, as there are many biologically relevant spiking behaviors to consider. To illustrate the richness of this open field, we selected two prototypical spiking types that we reported in a recent publication [13] and we explore in those circuits some basic effects of recursion. We shall consider only the excitatory case, which is relatively simpler as in the case of a LIF neuron always provides a converged self-consistent state.

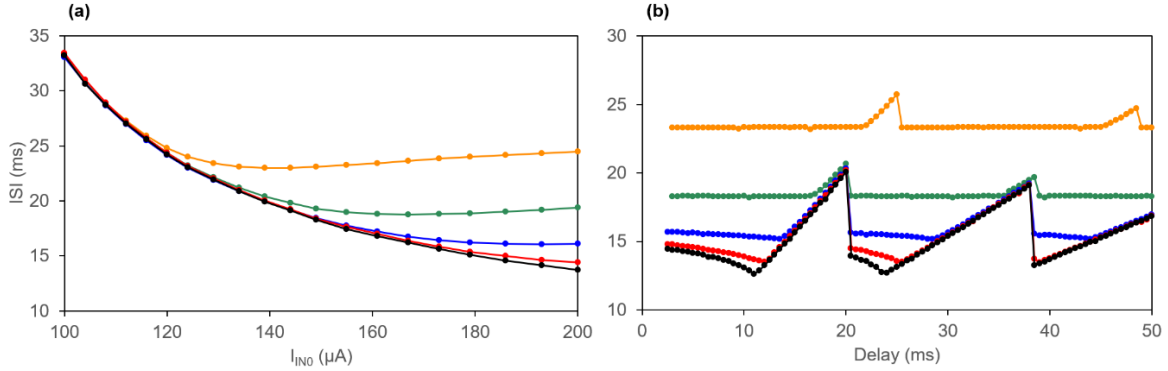
5.1 Autapse in a Spike Frequency Adaptation neuron type.

The Spike Frequency Adaptation (SFA) behavior is characterised by a neuron that under a constant applied input starts spiking at a fast rate, which progressively decreases towards a final lower value. This is the result of a second, so called "slow dynamical variable" [47]. We have implemented the circuit following our previous work [13] and supplemented it with an autapse as described in the Main Text. The circuit to implement this is shown in Fig. 8



Supplemental Figure 8. Circuit to implement the Spike Frequency Adaptation (grey), supplemented by the autapse feedback loop (yellow). Circuit component are $R_{6a}=20k\Omega$, $25 k\Omega$, $30 k\Omega$, $35k\Omega$, $R_{6b}=100k\Omega$, $R_7=100k\Omega$, $C_2=4.7\mu F$, $Q_4=ZVP3310$, $Q_5=ZVN0124$, Diode=1N4148. (The rest of components are as in section S1)

We first obtained the basic behavior (ie no autapse feedback) of the SFA circuit, that we show in Fig.9a

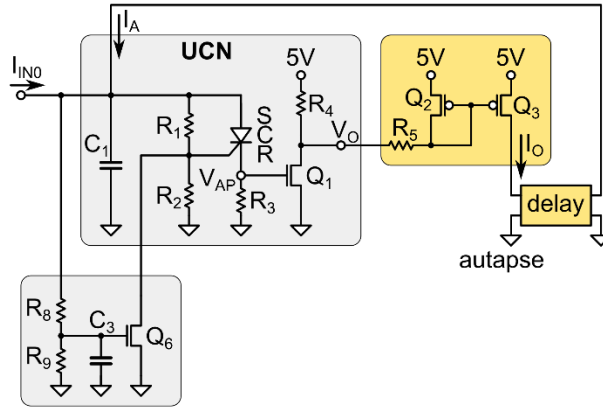


Supplemental Figure 9. (a) Basic SFA behavior without the autapse connection. Inter-Spike time Interval (ISI) which is the inverse of the instantaneous firing rate as a function of the constant current input I_{INO} . The value of ISI corresponds to the final low frequency rate after the initial adaptation transient time [13]. At zero adaptation (black) we have the LIF behavior. As adaptation is increased (red to orange) at fixed I_{INO} , we observe that the ISI increases or, equivalently, the adapted frequency becomes lower. (b) Self-consistent converged frequency of the SFA circuit as a function of the delay Δt for a fixed $I_{INO}=150\mu A$ and a gain $\beta=1$. The effect of the autapse at zero adaptation (black) corresponds to the LIF case discussed in the Main Text. The strength of the adaptation is controlled by the resistors R6a and R6b in the circuit of Fig.8. Zero adaptation (ie LIF behavior) corresponds to R6a = open circuit, then red to orange=35 k Ω , 30 k Ω , 25 k Ω , 20k Ω .

The degree of frequency adaptation is controlled by the resistors R6a and R6b in the circuit of Fig.8. At zero adaptation (R6a = open circuit) the neuron is the LIF model and the behavior of the ISI as function of Delay for a fixed input current are the same as in Fig. 2a of the Main Text, which shows a modulation of the ISI. As the degree of adaptation is increased (lowering the resistance R6a) the ISI increases and the modulation is increasingly suppressed. This indicates that the enhanced contribution to the I_{INO} via the autapse is eventually overwhelmed and becomes regulated by the adaptive feedback loop. This is an interesting situation where there are two feedbacks in competition, one is the excitatory feedback loop of the autapse, and the other is the feedback of the slow variable, which has an inhibitory effect (ie lowers the spiking rate). The small triangle modulation cannot be fully suppressed as it corresponds to the coincidence of the autapse feedback spike with the on-state of the thyristor, as we discussed in the Main Text.

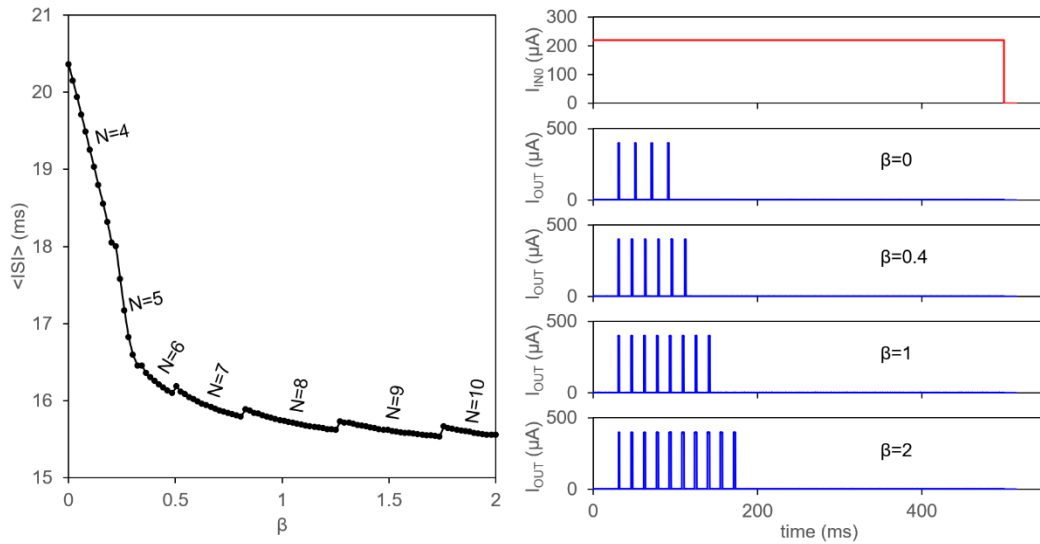
5.2 Autapse in a Phasic Burst neuron type.

We consider now a qualitatively different type of spiking behavior, that of Phasic Burst [13, 47]. In this case, as before, there is a constant applied current I_{INO} which, in the absence of autapse feedback produces a single train with a fixed number of spikes and then the spiking stops. We implemented this behavior with the circuit shown in Fig.10, which is similar to that introduced in [13].



Supplemental Figure 10. Circuit to implement the Phasic Burst spiking (grey), supplemented by the autapse feedback loop (yellow). $I_{INO}=220\mu A$ and $\Delta t=15\text{msec}$. Circuit component are $R_8=38k\Omega$, $R_9=62k\Omega$, $C_3=4.7\mu F$, $Q_6=ZVN0124$. (The rest of components are as in section S1)

The choice of the resistors R_8 and R_9 and the capacitor C_3 allows to set the time constant of the circuit block that controls the number of spikes in the train. This block simply stops the spiking by acting on the potential at the Gate of the SCR. For simplicity we chose the previously mentioned parameter values such that the phasic burst consists of a four-spike train (Fig.11). The interesting question is whether the excitatory feedback loop of the autapse may qualitatively change the behavior. In Fig. 11 we show the results for and increasing excitatory feedback parameter β .

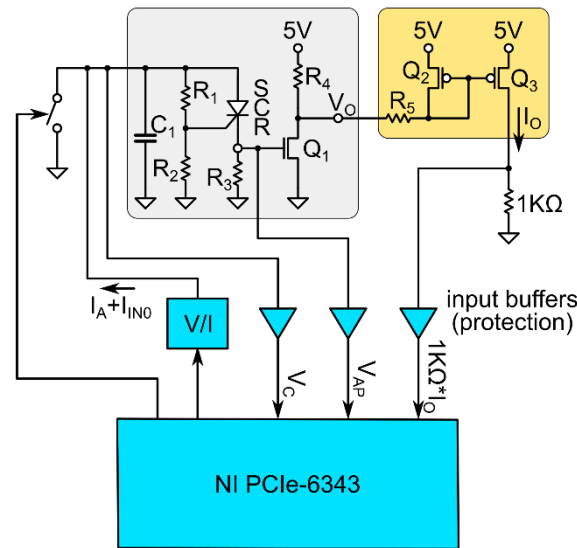


Supplemental Figure 11. Left: Average Inter-Spike time Interval (ISI) as a function of the excitatory feedback parameter β . The value N indicates the number of spikes in the burst. Right: The top panel shows the constant input current I_{INO} . Below, the case without autapse $\beta=0$, which shows a single 4-spike train. Lower panels illustrate the output when the excitatory autapse strength is increased.

It is interesting to observe in the figure that the effect of the autapse is to increase the number of spikes in the phasic train and also to increase the frequency rate. We can understand this result qualitatively by noticing that the effect of the feedback is to enhance the I_{INO} within the preset time window of the train, and in this way increase the number of emitted spikes. Thus the delay Δt (15 msec) has to be smaller than the time constant of the phasic block discussed before, which is 110 msec for the component values indicated in Fig.10.

6. Implementation of the Delay line and acquisition of signals

For this work, we need a delay line where the delay can be arbitrarily adjusted in the range $\sim 1\text{ms} - 50\text{ms}$ (e.g., see Fig. 2 in the Main Text). Additionally, we need to generate the input bias current and acquire various signals to monitor the circuit behavior. For these requirements, we implemented the setup presented in the figure below:



Supplementary Fig. 12. Experimental setup.

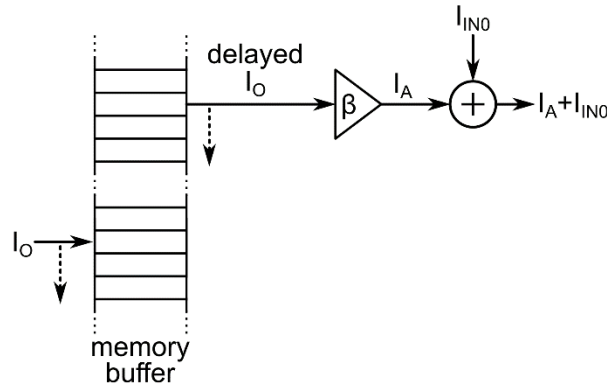
At the core of these setup there is a National Instruments NI PCIe-6343 data acquisition board (controlled by a LabVIEW-based control software). We use 3 analog voltage inputs (16 bits) and one analog voltage output of the NI PCIe-6343.

The analog inputs register V_C (voltage on the Membrane capacitor C_1), V_{AP} (the Action Potential signal on R_3), and $I_O \cdot 1K\Omega$ (to monitor Output current). Notice that, to protect the

acquisition board, we use input buffers based on TL084 op-amps. These signals are acquired at a rate of 100ksps.

The analog output generates $I_{IN} = I_{IN0} + I_A$ at a rate of 100ksps (e.g., see Fig. 5c in the Main Text). Notice that the output of the NI PCIe-6343 acquisition board is a voltage signal. Therefore, we use a circuit like the one proposed in Fig.7 of Ref. [30] to convert the voltage into a current.

The delay of the I_A signal is implemented by software as sketched in the figure below:



Supplementary Fig. 13. Implementation of the delay by software.

Each sample of I_O is sequentially store in a memory buffer at a rate of 100ksps. At the same rate, a delayed I_O is sequentially read from the same buffer, but from N memory positions before. This naturally creates a delay $\Delta t = N \cdot 10\mu s$. Then, the delayed I_O is multiplied by β and added to I_{IN0} to obtain $I_{IN} = I_{IN0} + I_A$.

Additionally, we implemented a reset switch using a IRLML6246TR NMOS transistor. This reset switch is not really necessary, but it was handy to have the V_C signal starting from 0V at each experiment (e.g., notice in Fig. 5d of the Main Text that V_C is hold at 0 during the first 10ms of the experiment).

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